

EXECUTIVE SUMMARY

OVERALL RISK LEVEL

HIGH

THREAT CLASSIFICATION

ZERO_DAY_ANOMALY

PREDICTION CONFIDENCE

99.9627352690299

Intrusion Detection Event Analysis

Attack Type: ZERO_DAY_ANOMALY

Severity: HIGH

Risk Score: 84.97

Confidence: 99.96%

1. Executive Summary

A high-severity **Zero-Day Anomaly** has been detected with near-certain confidence (99.96%). The risk score of **84.97** indicates a critical threat that could bypass traditional defenses, as it exploits an unknown vulnerability or novel attack vector. Immediate investigation and containment are required to prevent potential compromise of systems, data exfiltration, or lateral movement within the network.

2. Threat Analysis

- Attack Nature:** Zero-day anomalies indicate previously unidentified attack patterns, suggesting the use of an unpatched vulnerability or advanced evasion technique.
- Detection Basis:** The high confidence score implies strong behavioral or heuristic indicators (e.g., abnormal process execution, unusual network traffic, or privilege escalation attempts).
- Possible Tactics:** Could align with **Initial Access (TA0001)** or **Execution (TA0002)** in the MITRE ATT&CK framework, such as:
 - Exploitation of undocumented software flaws.
 - Fileless malware or living-off-the-land (LOLBin) techniques.
 - Polymorphic payloads evading signature-based detection.

3. Potential Impact

- Operational Disruption:** Unauthorized access could lead to system instability, ransomware deployment, or service outages.
- Data Breach Risk:** Exfiltration of sensitive data (PII, intellectual property, credentials).
- Lateral Movement:** Attackers may pivot to critical assets if not contained.
- Reputational & Compliance:** Regulatory penalties or loss of stakeholder trust if exploited.

4. Recommended Actions

Short-Term (Immediate Response):

- Isolate Affected Systems:** Quarantine endpoints/networks showing anomalous activity.
- Enhance Monitoring:** Deploy EDR/XDR tools to scrutinize process trees, memory, and network connections.
- Threat Hunting:** Search for related IOCs (unusual PowerShell/WMI usage, unexpected outbound connections).
- Patch & Mitigate:** Apply temporary mitigations (e.g., disabling vulnerable services, restricting permissions).

Long-Term (Remediation & Resilience):

- **Vulnerability Assessment:** Collaborate with vendors to identify/address the zero-day flaw.
- **Behavioral Detection Tuning:** Update SIEM/IDS rules to flag similar anomalies.
- **Incident Retrospective:** Conduct a root-cause analysis to improve defenses.
- **User Training:** Reinforce phishing/social engineering awareness (common zero-day delivery vectors).

Stakeholder Communication:

- **Leadership:** Brief on business risks and containment status.
- **IT Teams:** Share actionable IOCs and mitigation steps.
- **Legal/Compliance:** Assess reporting obligations (if data exposure is suspected).

Next Steps: Escalate to the incident response team for forensic analysis and coordinate with threat intelligence providers for emerging zero-day advisories.

Let me know if you require additional details or support.

— Senior SOC Analyst

INCIDENT INFORMATION

SOURCE IP	DEST IP	PROTOCOL	DETECTION TIME
192.168.1.164	192.168.56.1	6	2026-07-08T19:39:12.821739

AI EXPLAINABILITY (SHAP)

TOP CONTRIBUTING FEATURES		
packet_count		+2.1938
duration		-0.6220
total_bytes		-0.5555
std_packet_size		+0.4733
mean_packet_size		-0.2616
mean_iat		-0.1624

SYSTEM RESPONSE

Anomaly Detected

Sparse Autoencoder MSE: 2.621995052849362e-05

Classification Complete

LightGBM assigned label: ZERO_DAY_ANOMALY

Policy Evaluated

Severity HIGH processed.

Action Executed

DRIFT & ADAPTATION

CONCEPT DRIFT STATUS

STABLE

ADWIN DETECTION

No significant distribution shift detected in risk scores.

MODEL RETRAINING REQUIRED

False

TECHNICAL EVIDENCE

RISK SCORE	ANALYST OVERRIDE
84.96832497867543 / 100	Pending